A Semantic Modeling Approach for Medical Image Semantic Retrieval Using Hybrid Bayesian Networks

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Abstract

A multi-level semantic modeling method, which integrates Support Vector Machines (SVM) into hybrid Bayesian networks (HBN), is proposed in this paper. SVM discretizes the continuous variables of medical image features by classifying them into finite states as middle-level semantics. Based on the HBN, the semantic model for medical image semantic retrieval can be designed at multi-level semantics. To validate the method, a model is built to achieve automatic image annotation at the content level from a small set of astrocytona MRI (magnetic resonance imaging) samples. Multi-level annotation is a promising solution to enable medical image retrieval at different semantic levels. Experiment results show that this approach is very effective to enable multi-level interpretation of astrocytona MRI scan. It outperforms the Bayesian network-based model using k-nearest neighbor classifiers (K-NN). This study provides a novel way to bridge the gap between the high-level semantics and the low-level image features.

1. Introduction

With the rapid adoption of medical Picture Archival and Communication Systems (PACS) in hospitals, the demand for semantic image classification is becoming very high in order to support the effective medical image retrieval at the semantic level. For example, in tumor treatment planning, therapists are often interested in its malignant degree by comparing the retrieved historical cases using diagnostic image semantic features at the concept level. To generate the semantic image concepts, it is necessary to build a semantic model performing an automatic annotation of the medical images. Multi-level annotation of images is a promising solution to enable semantic retrieval by using various keywords at different semantic levels. On the other hand, a multi-level semantic model is able to represent medical images at the different content levels. Furthermore, a mature medical image semantic retrieval system would be helpful to assist doctors in diagnostics and treatment. The main task of a semantic model is to map the low-level image features into high-level semantics, bridging the gap between them.

Semantic modeling can be viewed as a pattern recognition task. Given a training set of images labeled with keywords that describe the image semantics, many state-of-the-art models performed the clustering merely based on the visual features, leading to a poor performance [1][2]. Recently, more models were introduced by applying machine learning techniques for automatic image annotation and retrieval, including the framework of multijects [3], a framework based on Bayesian networks (BN) with k-nearest neighbor (K-NN) classifiers [4] and multi-level annotation of natural scenes [5]. Naphade [3] proposed a Bayesian network-sbased multinet with multijects. The multinet was a multi-level semantic model and the

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multijects achieved middle semantics such as objects, sites or events. In case of static sites, Gaussion mixture models (GMM) were used, and in case of events and objects with spatio-temporal support, hidden Markov models (HMM) were applied. Jiebo Luo [4] used K-NN to model middle-level semantics and built a Bayesian networks-based framework for semantic image understanding. Fan [5] used a two-level model to associate salient objects and their corresponding keywords. At the first level, salient objects were automatically extracted and classified by support vector machines (SVM). At the second level, Gaussian Mixture Models learned by an adaptive EM (Expectation Maximum) technology was used to annotate images. But all these methods required a large training set. In medical image diagnosis, collecting a large training sample set with various level of labels is very difficult. In this case, a pattern recognition approach suitable for a small training set is necessary. SVM has shown its capacity in pattern recognition and recently been used for content-based image retrieval [6,7,8], which is more suitable for a small training set. But all of them were single level visual models. In medical image diagnosis, modeling image semantics often needs a multi-level semantic concept expression at various content levels. For example, when modeling tumor malignant degrees, it needs to represent tumor characters described by other concepts rather than the tumor malignant degree. To use statistical machine learning to build a computational framework for mapping low-level image features to middle and high-level semantics, a hybrid Bayesian network-based semantic model is proposed in this paper, which integrates SVM into BN. SVM discretizes the continuous variables and bridges the low-level image features with the middle semantics together simultaneously. It allows the model to handle continuous inputs in classical BN. We apply this method in modeling semantic classification of astrocytoma malignant degree, by fusing middle level semantics with low-level image features. In addition, the model can also perform probabilistic predictions of astrocytoma malignant levels. Experiment results demonstrate that our method is very effective to establish multi-level annotation of astrocytoma MRI from a small set of training samples, which outperforms those models using K-NN.

2. Bayesian Networks

BN are directed, acyclic graphs that encode the cause-effect and conditional independence relationships among variables in the probabilistic reasoning system [9], where nodes of the graphs are the variables of the domain one wants to model. The links between the nodes represent the conditional probabilities of inferring the existence of one variable (the destination of the link). The direction of the edge goes from a parent node to a child node indicting a conditional dependency of a child given the parent.

Bayesian networks have been proved to be an effective knowledge representation and inference engine in artificial intelligence and expert systems. Domain-specific knowledge can be incorporated in the network structure, and a complicated joint probability distribution can be reduced to a set of conditionally independent relationships that are easier to characterize. Thus, a Bayesian network can represent the dependence between random variables (features) and give a simple and reasonable specification of the joint probability distribution for a domain.

According to Bayesian rule, the posterior probability can be expressed by the joint probability, which can be further expressed by the conditional and prior probabilities. In a BN, when it fuses many sensors, the expended Bayesian rule is applied as

$$P(Hi \mid \vec{e}) = \frac{P(\vec{e} \mid Hi)P(Hi)}{\sum_{i=1}^{s} P(\vec{e} \mid Hi)P(Hi)} \quad (1)$$

Where Hi denotes astrocytoma states and $\vec{e}$ denotes observed features.

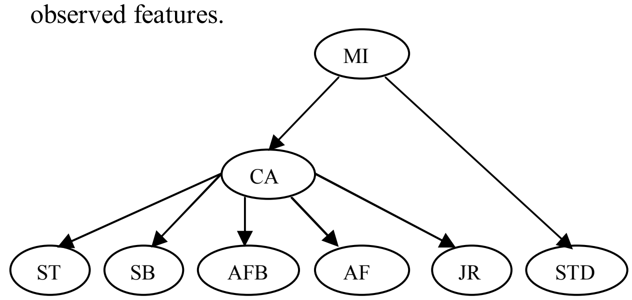

Figure.1 A segment of a BN for diagnosing myocardial infarction

As an example, Fig.1 shows a BN example for diagnosing myocardial infarction [10], where leaf nodes denote observed symptoms and the output node denotes diseases. All nodes are discrete and the posterior probability P (MI/ symptoms) gives the diagnosis belief.

3. Hybrid Bayesian network fusing SVM

A good semantic model should be able to achieve a middle-level and high-level understanding of the semantics of the image contents. For example, a semantic framework, which models astrocytoma malignant degree, should include multi-level understanding of the images. For examples, at the

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middle level it should understand the tumor uniformity, the enhanced contrast etc, and at the high level the malignant degree. SVM can be used to extract these middle-level semantics, as nodes in Bayesian network, from low-level image features. Since nodes in the classical BN are discrete, they are not able to accept continuous inputs, such as medical image low-level features. SVM can map the continuous inputs to discrete nodes with finite states. K-NN as shown in [4] can do it also, but it can only achieve satisfying results when applied with a large training set.

In this network, like a concept detector, SVM is used to discretize continuous inputs and map low-level image features to middle semantics simultaneously. In the end, a BN is used to combine these middle-level semantics. This BN is called BN-SVM, defined as

$$\text{BN-SVM}=(X, \text{SVM}(\hat{Y}), \hat{Y}, L, P) \qquad (2)$$

Where X stands for a set of discrete nodes. $\hat{Y}$ stands for a set of continuous nodes. SVM ($\hat{Y}$) stands for a set of nodes defined by discretizing $\hat{Y}$. L contains the set of directed links, and P is the joint probabilistic distribution of Bayesian networks.

In this study, we select the radial basis function (RBF), $k(X_i, X_j) = \exp(-\|X_i-X_j\|^2/\sigma^2)$.

We apply BN-SVM to build an intelligent semantic model to identify astrocytoma malignant degree.

To evaluate our model, precision ρ and recall Q are used to measure the performance.

$$\rho=\eta/(\eta+\varepsilon) \qquad Q=\eta/(\eta+\theta). \qquad (3)$$

Where η is the set of true positive samples that are related to the corresponding semantic type and are detected correctly, ε is the set of true negative samples that are irrelevant to the corresponding semantics but are detected incorrectly, and θ is the set of false positive samples that are related to the corresponding semantic type but are mis-detected.

4. The BN-SVM-based semantic model for astrocytoma malignant degrees

4.1 Model structure and probability learning

The model structure is designed according to medical expert knowledge and general principles of BN, as shown in Fig.2. Where

1) The node 'malignant degree' is output node, with two states: 1- benign; 2- malignant.
2) The round nodes denote continuous variables representing low-level image features extracted directly from patients' MRI by utilizing medical image processing technology. Their parent nodes (SVM) denote their corresponding discrete inputs. The SVM outputs specify their middle semantics.

We trained the BN with the ground truth from pathological classification results and low-level image features from MRI. By using EM technology [9], a probability model expressed by the conditional probability table (CPT) is built.

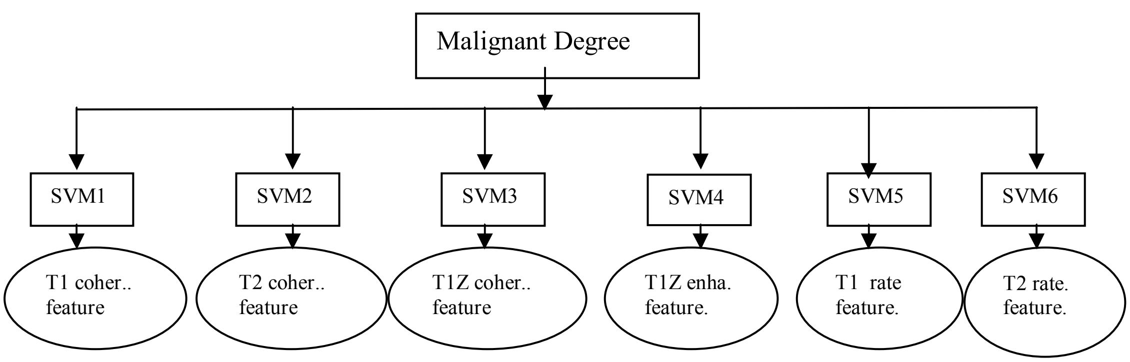

Figure 2 A BN-SVM-based semantic model for astrocytoma malignant

4.2 Inference of the semantic BN-SVM model

After the CPT of the model is obtained, the hybrid streams of low-level and semantics from test samples are piped into the BN-SVM-based inference engine, then the posterior probabilities of the benign and malignant state are rendered respectively. In the end, the state having the maximal posterior probability is the diagnosis output of the test sample. Fig.3 gives the inference flow chart.

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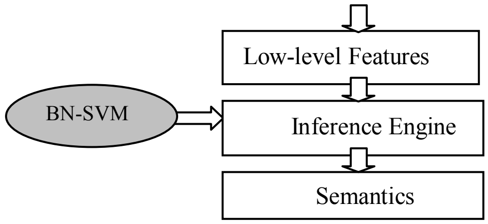

Figure 3 Inference flow of BN-SVM

4.3. Knowledge expression of the BN-SVM

From the knowledge engineering point of view, a BN is a graphical model. The structure of the network is formulated in a graphical communication language. It consists of a qualitative part, where features from graph theory are used, and a quantitative part with potentials, which are real-valued functions over sets of nodes from the graph. The graphical part specifies the kind of potentials and their domains. Fig.5 shows the graphical knowledge expression when applying the modeling process to the sample shown in Fig.4.

5. Experiment results

All sample images were acquired from a Philips 1.5 Tesla MRI equipment with a standard head coil. Image size is 256x256. The data collection consists of 124 astrocytoma samples (3 MR images per sample, including T1, T2 and T1Z weighted image) with pathological classification results. Among them, 70 samples (35 benign and 35 malignant) are used as training set and the rest 54 samples (21 benign and 33 malignant) are used for the test purpose. Prior probability was evaluated by using frequency counting from the 70 training samples. Once the BN-SVM has been constructed and trained, it can be used to infer the posterior probability for the test samples very efficiently. In term of the maximal posterior probability rule, different level semantics of the samples are decided, and added as their annotation. The annotations consist of six relevant middle-level semantic concepts and high-level semantics.

5.1 Extracting middle-level semantics using SVM

In this experiment, we used six SVM to extract middle-level semantics 'T1 uniformity', 'T2 uniformity', 'T1Z uniformity', 'T1Z enhanced Contrast ', 'T1' and 'T2' respectively as shown in Fig.2. The detection performance is shown in Table 1.

Table 1 Detection performance of six middle-level semantics based on SVM

Semantics	Optimal parameters	Accuracy
T1 uniformity	σ=0.8, C=1000	96.3%
T2 uniformity	σ=0.8 , C=1000	98.0%
T1Z uniformity	σ=0.8 , C=1000	98.0%
T1Z enhan. Contrast	σ=0.03, C=1000	98.0%
T1	σ=0.08, C=1000	94.4%
T2	σ=0.08, C=1000	94.4%

5.2 Extracting high-level semantics using BN-SVM

After extracting six middle-level semantics, a BN is used to achieve high-level semantics 'malignant degree'. To evaluate the model, after 15 iterations, and comparison with pathological classification results, the final precision, recall and false alarm rate reflecting the semantics of 'malignant' and 'benign' are shown in Table 2, where one can also found the similar result by a BN using K-NN to discretize continuous inputs rather than SVM, which is called BN-KNN. It can be found out that using BN-SVM it has improved the recall rate that is very important in medical diagnosis. In BN-SVM approach, there are 8 malignant samples are annotated to benign samples by error, and this is the reason of 27.6% false alarm rate and 72.7% recall of semantics "malignant", so precision of semantics "benign" reaches a poor 72.4%. An ideal image annotation system would have a high annotation recall and annotation precision simultaneously. By all appearances, the recall for semantics 'malignant' and 'benign' of BN-SVM is higher than BN-KNN.

As an example, the inference result of a malignant astrocytoma sample in Fig.4 is that the probability of benign state is 0.0802, and the probability of malignant state is 0.9198. The classification result is malignant in terms of the maximal posterior probability. The knowledge expression of the model is shown in Fig.5. At the content level, Fig.5 describes the two level semantic characters. It gives the tumor's middle level semantics that asymmetrical T1 distribution, asymmetrical T2 distribution, asymmetrical T1Z distribution as well as high contrast enhancement. And it also shows both longer T1 and T2 value than normal tissue. Finally, this model gives the posterior probabilities of outputs semantics.

Some T1 weighted images of the samples annotated as "malignant astrocytoma" are shown in Fig.6.

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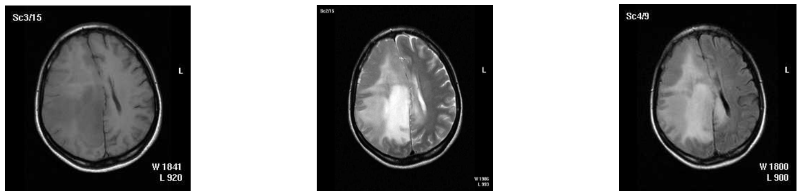

(a) **T1 weighted image** (b) **T2 weighted image** (c) **T1Z weighted image**

Figure. 4 A malignant astrocytoma sample

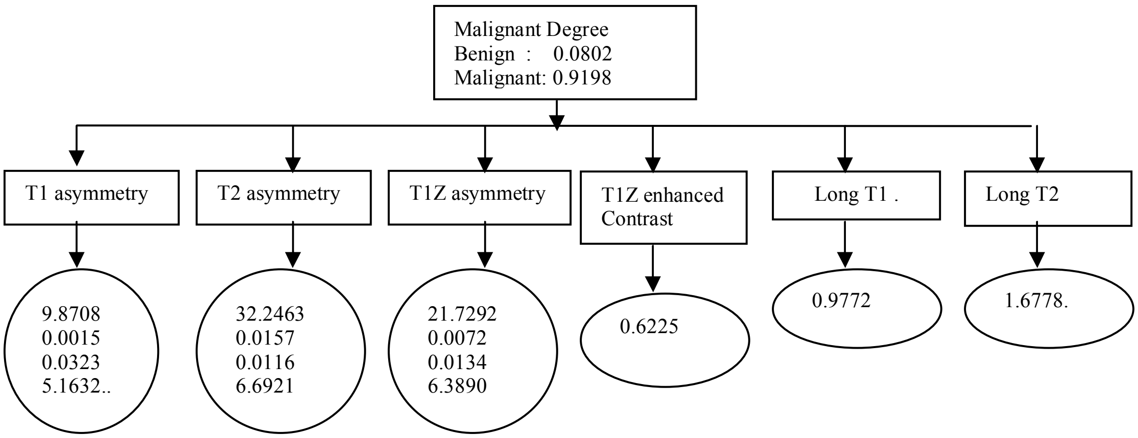

Figure. 5 A BN-SVM-based semantic expression for malignant level of the sample in Fig.4

Table 2. Semantics classification performance comparison between BN-SVM and BN-KNN

Model	Semantics	*Precision*	*Recall*	*False Alarm*
BN-SVM	*Benign*	72.4%	100%	27.6%
	Malignant	100%	72.7%	0%
BN-KNN	*Benign*	92.9%	61.9%	7.1%
	Malignant	100%	21.2%	0%

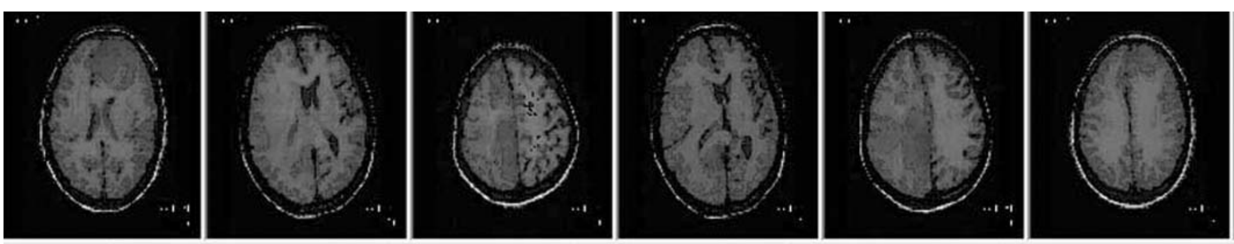

Figure 6 T1 weighted images of the samples annotated as "malignant astrocytoma"

6. Conclusions

This paper proposes a multi-level semantic modeling method using SVM and BN. The SVM discretize the continuous variables into discrete ones and map low-level image features to middle semantics simultaneously, and then are embedded into a BN, so that a multi-level semantic model is built. In another word, the model classifies the image into multi-level semantic concepts via concept hierarchy. It enables more effective image retrieval so that users can have more flexibility to specify their query by using various keywords at different semantic levels. We applied the method to semantic modeling of astrocytoma malignant degrees. Experiment results show the efficiency of our new framework. It outperforms the BN-KNN, a BN-based model using K-NN to extract middle semantics.

Compared with the model using K-NN as classifiers [4], our BN-SVM model has higher recall that is very important to medical image retrieval. Meanwhile, K-NN only fits for a large training set.

Collecting such a large labeled medical sample set is rather a difficult job. For a small training set, our approach behaves more efficiently. Therefore, it fits better for medical image semantic modeling.

Automatic learning algorithm utilizing machine learning is highly dependent on the samples. Whether samples are enough or not is concerned with the sample distribution and recognition tasks, so how to choose samples is very important. In our experiment, the train-ing samples are collected to cover typical astrocy-toma cases as much as available. It can be further improved in terms of filtering image features and enh-ancing their precision. Handpicked samples and increased number of trials would also improve the accuracy.

Unlike [11], which utilized Neural Networks (NN) and modeled malignant degree of astrocytoma at single-level semantic concept, NN tends to memorize the training set without domain knowledge, and automatic algorithms alone may over fit the data by focusing on spurious correlation as opposed to true causality among cues. Generally, it needs more training data to achieve the same result. Many studies showed that the domain knowledge is critical for training a network that exhibits good generalization. On the other hand, NN is treated as a black box that is incomprehensible, and it didn't enable semantic image retrieval at different semantic levels.

On the same line, Bayesian networks combine domain knowledge and automatic algorithms to discover the model from data [9] as an intelligent tool. It can avoid over fitting of data and has high accuracy even without enough training data. BN is also an effective knowledge representation with reasonable semantics on every node. Different from other data analysis tools, such as decision trees and NN, Bayesian networks have the valuable property that one can query any part of their knowledge. Our novel modeling approach shown in this paper can represents different level of features more rationally and can handle continuous inputs, so it fits better for medical image diagnosis.

An important step forwards would be to extend the knowledge and network inference capacity of the model by applying more training samples, so that the accuracy will be increased. At the same time, the model should be applied to clinic in order to get feedbacks and gain further improvement.

Acknowledgments

The authors acknowledge the kindly supports of China NNSF(Grant No: 60472063).

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